Natural Language Processing (NLP) with Python

Table of Contents:

1. What is Natural Language Processing?
2. Applications of NLP
3. Understanding Natural Language Processing (NLP)
4. Rule-based NLP vs. Statistical NLP
5. Components of Natural Language Processing (NLP)
6. Current challenges in NLP
7. Easy to Use NLP Libraries
8. Exploring Features of NLTK
9. Word Cloud
10. Synonym, antonym, definition
11. Stemming
12. Lemmatization
13. Part-of-Speech (PoS) tagging

Exploring Features of NLTK:

Synonyms, Antonyms and Definitions

Get synonyms from WordNet

If you remember, we installed NLTK packages using nltk.download(). One of the packages was WordNet.

WordNet is a database that is built for natural language processing. It includes groups of synonyms and a brief definition.

Synset is a special kind of a simple interface that is present in NLTK to look up words in WordNet. Synset instances are the groupings of synonymous words that express the same concept.

You can get these definitions and examples for a given word like this:

```
from nltk.corpus import wordnet

syn = wordnet.synsets("accurate")

print(syn[0].definition())
```

```
print(syn[0].examples())
```

```
conforming exactly or almost exactly to fact or to a standard or
performing with total accuracy
['an accurate reproduction', 'the accounting was accurate', 'accurate
measurements', 'an accurate scale']
```

You can use WordNet to get synonymous words like this:

```
from nltk.corpus import wordnet
synonyms = []
for syn in wordnet.synsets('Computer'):
    for lemma in syn.lemmas():
        synonyms.append(lemma.name())
print(synonyms)
```

```
['computer', 'computing_machine', 'computing_device', 'data_processor',
'electronic_computer', 'information_processing_system', 'calculator',
'reckoner', 'figurer', 'estimator', 'computer']
```

Get antonyms from WordNet

You can get the antonyms words the same way, all you have to do is to check the lemmas before adding them to the array if it’s an antonym or not.

```
from nltk.corpus import wordnet

antonyms = []

for syn in wordnet.synsets("small"):

    for l in syn.lemmas():

        if l.antonyms():

            antonyms.append(l.antonyms()[0].name())

print(antonyms)
```

```
['large', 'big', 'big']
```

The NLTK wordnet has another great feature, by using it we can check whether two words are nearly equal or not. It will return the similarity ratio from a pair of words.

The wup_similarity method is short for Wu-Palmer Similarity, which is a scoring method based on how similar the word senses are and where the Synsets occur relative to each other in the hypernym tree.

```
import nltk
from nltk.corpus import wordnet     #Import wordnet from the NLTK
first_word = wordnet.synset("Travel.v.01")
second_word = wordnet.synset("Walk.v.01")
print('Similarity: ' + str(first_word.wup_similarity(second_word)))
first_word = wordnet.synset("Good.n.01")
second_word = wordnet.synset("zebra.n.01")
print('Similarity: ' + str(first_word.wup_similarity(second_word)))
```

```
Similarity: 0.6666666666666666
Similarity: 0.09090909090909091
```

Stemming:

We use Stemming to normalize words. In English and many other languages, a single word can take multiple forms depending upon context used. For instance, the verb “study” can take many forms like “studies,” “studying,” “studied,” and others, depending on its context. When we tokenize words, an interpreter considers these input words as different words even though their underlying meaning is the same. Moreover, as we know that NLP is about analyzing the meaning of content, to resolve this problem, we use stemming.

Stemming normalizes the word by truncating the word to its stem word. For example, the words “studies,” “studied,” “studying” will be reduced to **“studi,”** making all these word forms to refer to only one token. Notice that stemming may not give us a dictionary, grammatical word for a particular set of words.

Let’s take an example:

a. Porter’s Stemmer Example 1:

In the code snippet below, we show that all the words truncate to their stem words. However, notice that the stemmed word is not a dictionary word.

#stemming Example

#import stemming Library

```
from nltk.stem import PorterStemmer

porter = PorterStemmer()
# word-list for stemming
word_list =["Study","Studying","Studies","Studied"]
for w in word_list:
   print(porter.stem(w))
```

```
studi
studi
studi
studi
```

b. Porter's Stemmer Example 2:

In the code snippet below, many of the words after stemming did not end up being a recognizable dictionary word.

```
#stemming Example
#import stemming Library
from nltk.stem import PorterStemmer
porter = PorterStemmer()
# word-list for stemming
word_list =["Studies","leaves","decreases","plays"]
for w in word_list:
   print(porter.stem(w))
```

```
studi
leav
decreas
play
```

c. SnowballStemmer:

SnowballStemmer generates the same output as porter stemmer, but it supports many more languages.

```
#stemming Example
#import stemming Library
from nltk.stem import SnowballStemmer
snowball = SnowballStemmer("english")
# word-list for stemming
word_list =["Study","Studying","decreases","plays"]
```

```
for w in word_list:
    print(snowball.stem(w))
```

```
studi
studi
decreas
play
```

d. Languages supported by snowball stemmer:

```
#stemming Example
#import stemming Library
from nltk.stem import SnowballStemmer

#print languages supported
SnowballStemmer.languages
```

```
('arabic',
 'danish',
 'dutch',
 'english',
 'finnish',
 'french',
 'german',
 'hungarian',
 'italian',
 'norwegian',
 'porter',
 'portuguese',
 'romanian',
 'russian',
 'spanish',
 'swedish')
```

Various Stemming Algorithms:

a. **Porter's Stemmer:**
   Advantage: It produces the best output and it has lowest error rate
   Limitation : The reduced words are not always the real words

b. **Lovin's Stemmer:**
   Advantage: Its fast and it can handle irregular plurals like foot - >feet
   Limitation: Its very time consuming and it has higher error rate

c. **Dawson's Stemmer:**
   Advantage: Its fast and covers more suffixes
   Limitation: Its bit complex to implement

d. **Krovetz Stemmer:**

Advantage: Its light in nature and used as a pre-stemmer to other stemmers
Limitation: it is inefficient in case of larger documents

e. **Xerox Stemmer:**

Advantage: It works fine with larger documents and the error rate is low
Limitation: it may result in overstemming and it is language dependent

f. **Snowball Stemmer:**

Advantage: It supports more languages
Limitation: The reduced words are not always the real words

Lemmatization:

Lemmatization tries to achieve a similar base "stem" for a word. However, what makes it different is that it finds the dictionary word instead of truncating the original word. Stemming does not consider the context of the word. That is why it generates results faster, but it is less accurate than lemmatization.

If accuracy is not the project's final goal, then stemming is an appropriate approach. If higher accuracy is crucial and the project is not on a tight deadline, then the best option is amortization (Lemmatization has a lower processing speed, compared to stemming).

Lemmatization takes into account Part Of Speech (POS) values. Also, lemmatization may generate different outputs for different values of POS. We generally have four choices for POS:

Verb (v)
Examples : Study, Play, Learn, Am, Is, Are...
Noun (n)
Examples : Doctor, Engineer, Farm, Physiotherapist,Towards AI...
Adjective (a)
Examples : Beautiful, Elegant, Angry, Polite, Repulsive...
Adverb (r)
Examples : Badly, Slowly, Peacefully, Very, Extremely, Occasionally...

Figure 46: Part of Speech (POS) values in lemmatization.

Difference between Stemmer and Lemmatizer:

a. Stemming:

Notice how on stemming, the word “studies” gets truncated to “studi.”

```
#stemming Example
#import stemming Library
from nltk.stem import PorterStemmer
stemmer = PorterStemmer()
print(stemmer.stem('studies'))
```

```
studi
```

b. Lemmatizing:

During lemmatization, the word “studies” displays its dictionary word “study.”

```
#stemming Example
#import stemming Library
from nltk.stem import WordNetLemmatizer
lem = WordNetLemmatizer()
print(lem.lemmatize('studies'))
```

```
study
```

Python Implementation:

a. A basic example demonstrating how a lemmatizer works

In the following example, we are taking the PoS tag as “verb,” and when we apply the lemmatization rules, it gives us dictionary words instead of truncating the original word:

```
#stemming Example
#import stemming Library
from nltk.stem import WordNetLemmatizer
lem = WordNetLemmatizer()
word_list =['study','studying','studies','studied']
for w in word_list:
    print(lem.lemmatize(w,pos='v'))
```

```
study
study
study
study
```

b. Lemmatizer with default PoS value

The default value of PoS in lemmatization is a noun(n). In the following example, we can see that it’s generating dictionary words:

```
#stemming Example
#import stemming Library
from nltk.stem import WordNetLemmatizer
lem = WordNetLemmatizer()
word_list =['studies','leaves','decreases','plays']
for w in word_list:
    print(lem.lemmatize(w))
```

```
study
leaf
decrease
play
```

c. Another example demonstrating the power of lemmatizer

```
#stemming Example
#import stemming Library
from nltk.stem import WordNetLemmatizer
lem = WordNetLemmatizer()

word_list =["am","is","are","was","were"]
for w in word_list:
    print(lem.lemmatize(w, pos ="v"))
```

```
be
be
be
be
be
```

d. Lemmatizer with different POS values

```
#stemming Example
#import stemming Library
from nltk.stem import WordNetLemmatizer
lem = WordNetLemmatizer()
print(lem.lemmatize("studying", pos ="v"))
print(lem.lemmatize("studying", pos ="n"))
print(lem.lemmatize("studying", pos ="a"))
print(lem.lemmatize("studying", pos ="r"))
```

```
study
studying
studying
studying
```

Part of Speech Tagging (PoS tagging):

Why do we need Part of Speech (POS)?

Can you help me with the can?

Figure 53: Sentence example, “can you help me with the can?”

Parts of speech (PoS) tagging is crucial for syntactic and semantic analysis. Therefore, for something like the sentence above, the word “can” has several semantic meanings. The first “can” is used for question formation. The second “can” at the end of the sentence is used to represent a container. The first “can” is a verb, and the second “can” is a noun. Giving the word a specific meaning allows the program to handle it correctly in both semantic and syntactic analysis.

Python Implementation:

a. A simple example demonstrating PoS tagging.

```
#Pos tagging
import nltk
tag = nltk.pos_tag(["Studying","Study"])
print(tag)
```

```
[('Studying', 'VBG'), ('Study', 'NN')]
```

b. A full example demonstrating the use of PoS tagging.

```
#Pos tagging Exa
from nltk import sent_tokenize
from nltk import word_tokenize
sentence ="A very beautiful young lady is walking on the beach"

#Tokenizing words
token_words = word_tokenize(sentence)

for words in token_words:
   tagged_words = nltk.pos_tag(token_words)

print(tagged_words)
```

```
[('A', 'DT'), ('very', 'RB'), ('beutiful', 'JJ'), ('young', 'JJ'),
('lady', 'NN'), ('is', 'VBZ'), ('walking', 'VBG'), ('on', 'IN'),
('the', 'DT'), ('beach', 'NN')]
```

####### ##### ####

The above solution is a list of tuples

How to extract Nouns from the sentence (the list of tuples)

```
Output = [word for word in tagged_words
        if (word[1] == 'NN')]
Output
```

```
[('lady', 'NN'), ('beach', 'NN')]
```

```
If we want only the first value of the tuple then write the code as
```

```
Output = [word[0] for word in tagged_words
        if (word[1] == 'NN')]

print(Output)
```

```
['lady', 'beach']
```

Defining our own stopwords

```
words_no_punc = ["Heloo","good"," mroning","how","dogs","and","you",
"are","a","cat"]
stopwords =['dogs','cat']
clean_words =[]

for w in words_no_punc:
    if w not in stopwords:
        clean_words.append(w)

print(clean_words)
print("\n")
print(len(clean_words))
```

```
['Heloo', 'good', ' mroning', 'how', 'and', 'you', 'are', 'a']
```

Example of POS Tagging in NLTK

In the below example, we first tokenize the text and pass the tokens to NLTK pos_tag() function.

```
from nltk import pos_tag
from nltk import word_tokenize

text = "The way to get started is to quit talking and begin doing."
tokenizer = word_tokenize(text)
pos_tag(tokenizer)
```

```
[('The', 'DT'),
 ('way', 'NN'),
 ('to', 'TO'),
 ('get', 'VB'),
 ('started', 'VBN'),
 ('is', 'VBZ'),
 ('to', 'TO'),
 ('quit', 'VB'),
 ('talking', 'VBG'),
 ('and', 'CC'),
 ('begin', 'VB'),
 ('doing', 'VBG'),
 ('.', '.')]
```

Default Tagger in NLTK

NLTK has DefaultTagger function that is used to assign the default tag to the tokens. Let us see this with the help of an example.

Below, we first tokenize the text and then create an instance of DefaultTagger by adding the desired default tag ‘AD’. Finally, we pass the tokenized text to the DefaultTagger instance.

```
from nltk.tag import DefaultTagger

text = "The way to get started is to quit talking and begin doing."
tokenizer = word_tokenize(text)
tagging = DefaultTagger('Ad')

print(tagging.tag(tokenizer))
```

```
[('The', 'Ad'), ('way', 'Ad'), ('to', 'Ad'), ('get', 'Ad'), ('started',
'Ad'), ('is', 'Ad'), ('to', 'Ad'), ('quit', 'Ad'), ('talking', 'Ad'),
('and', 'Ad'), ('begin', 'Ad'), ('doing', 'Ad'), ('.', 'Ad')]
```

The contents below are for reference only

You can refer the below parts of speech – but no need to remember the contents

Below, please find a list of Part of Speech (PoS) tags with their respective examples:

1. CC: Coordinating Conjunction

(CC) Coordinating Conjunction
Words
And, But, For, Nor, Or, So, Yet...
Example
You can eat your cake with a spoon or fork.

2. CD: Cardinal Digit

(CD) Cardinal Digit
Words
One, Two, Three, Four, Five, Six...
Example
There are five flowers in the flower vase.

3. DT: Determiner

(DT) Determiner
Definite Article
The
Indefinite Article
A, An
Demonstratives
This, That, These, Those
Pronouns and Possessive Determiners
My, Your, His, Her, It's, Our, Their
Quantifiers
Any, A few, A little, Much, Many, Most, Some
Numbers
One, Ten, Thirty
Distributives
All, Both, Half, Either, Neither, Each, Every
Difference Words
Other, Another
Pre-determiners
Such, What, Rather, Quite

4. EX: Existential There

(EX) Existential There
Words
There
Example
There is a pen on the desk.

5. FW: Foreign Word

(FW) Foreign Words
Words
dolce, ersatz, esprit, quo, maitre
Example
They were selling ersatz products.

6. IN: Preposition / Subordinating Conjunction

(IN) Preposition / Subordinating Conjunction
Words
At, In, On, Above, Behind
Example
All speak at the same time.

7. JJ: Adjective

(JJ) Adjective
Words
New, Old, High, Special, Big, Local
Example
They live in a beautiful house.

8. JJR: Adjective, Comparative

(JJR) Adjective, Comparative
Words
Newer, Bigger, Higher
Example
The new bird is angrier than Robin.

9. JJS: Adjective, Superlative

(JJR) Adjective, Superlative
Words
Biggest, Darkest, Most
Example
Jupyter is the biggest planet in our solar system.

10. LS: List Marker

(LS) List Marker
Words
1), 2), 3)...
Example
1) One 2) Two 3) Three

11. MD: Modal

(MD) Modal
Words
Could, Will, Should
Example
Should we go outside in this situation?

12. NN: Noun, Singular

(NN) Noun , Singular
Words
Year, Home, Cost, Time, Education
Example
Why don't we stop by my house for coffee?

13. NNS: Noun, Plural

(NNS) Noun , Plural
Words
Pens, Books, Hats
Example
There are a bunch of pens on the table.

14. NNP: Proper Noun, Singular

(NNP) Proper Noun , Singular
Words
April, Africa, Pratik
Example
April is one of the hottest months.

15. NNPS: Proper Noun, Plural

(NNPS) Proper Noun , Plural
Words
Americans, Indians
Example
Most Americans are young at heart.

16. PDT: Predeterminer

(PDT) Predeterminer
Words
All, Both, Half
Example
They all were accepted.

17. POS: Possessive Endings

(POS) Possessive Endings
Words
Parent's, Pratik's
Example
It's Jan's assignment.

18. PRP: Personal Pronoun

(PRP) Personal Pronoun
Words
I, He, She
Example
She is very brilliant.

19. PRP$: Possessive Pronoun

(PRP$) Possessive Pronoun
Words
My, His, Her, Hers, Yours
Example
This is my book.

20. RB: Adverb

(RB) Adverb
Words
Really, Already, Still, Early, Now
Example
We are still waiting for them.

21. RBR: Adverb, Comparative

(RBR) Adverb, Comparative
Words
Worse, Less, Better
Example
This is way better than we anticipated.

22. RBS: Adverb, Superlative

(RBS) Adverb, Superlative
Words
Worst, Least, Best
Example
This is the best day ever.

23. RP: Particle

(RP) Adverb, Superlative
Words
Aboard, About, Across, Along, At, Away
Example
They were about to leave.

24. TO: To

(TO) To
Words
To
Example
We were going to the store.

25. UH: Interjection

(UH) Interjection
Words
Amen, Huh, Howdy, Dammit, Heck, Anyways
Example
What the heck!

26. VB: Verb, Base Form

(VB) Verb, Base Form
Words
Take, Play, Sit, Listen
Example
Let us play a game.

27. VBD: Verb, Past Tense

(VBD) Verb, Past Tense
Words
Took, Studied, Played
Example
We studied for an hour.

28. VBG: Verb, Present Participle

(VBG) Verb, Present Participle
Words
Playing, Studying
Example
They have been playing since morning.

.

29. VBN: Verb, Past Participle

(VBN) Verb, Past Participle
Words
Taken, Languished, Dilapidated
Example
They were taken hostages.

30. VBP: Verb, Present Tense, Not Third Person Singular

(VBP) Verb, Present Tense, Not 3rd Person
Words
Terminate, Appear, Cure
Example
We are going to terminate the contract.

31. VBZ: Verb, Present Tense, Third Person Singular

(VBP) Verb, Present Tense, 3rd Person
Words
Uses, Speaks, Slaps
Example
He generally speaks about sports.

32. WDT: Wh — Determiner

(WDT) Wh - Determiner
Words
That, What, Whatever, Which, Whichever
Example
What is your name?

33. WP: Wh — Pronoun

(WP) Wh - Pronoun
Words
Who, Whom, Which, What
Example
Who is that guy in the limo?

34. WP$: Possessive Wh — Pronoun

(WP$) Possessive Wh - Pronoun
Words
Whose
Example
Whose pen is this?

35. WRB: Wh — Adverb

(WRB) Wh - Adverb
Words
How, However, Why, Where, When
Example
How is this possible?

List of POS Tags in NLTK

Usually, in schools, we are taught about 9 different types of parts of speech – noun, verb, adverb, article, preposition, pronoun, adjective, conjunction, and interjection. But NLTK actually provides many categories and sub-categories of tags than just the traditional nine.

We can generate all the available POS tags by using nltk.help.upenn_tagset() function.

Below is the complete list of NLTK POS tags –

	Pos_tag	tag_name	example
0	$	dollar	[, –, –, —, A, A, C, HK, HK, M, NZ, NZ, S, U.S., U.S, …
1	”	closing quotation mark	[‘, ”]
2	(	opening parenthesis	[(, [, {]
3	)	closing parenthesis	[),], }]
4	,	comma	[,]
5	—	dash	[–]
6	.	sentence terminator	[., !, ?]
7	:	colon or ellipsis	[:, ;, …]
8	CC	conjunction, coordinating	[&, ‘n, and, both, but, either, et, for, less,…
9	CD	numeral, cardinal	[mid-1890, nine-thirty, forty-two, one-tenth, …
10	DT	determiner	[all, an, another, any, both, del, each, eithe…
11	EX	existential there	[there]
12	FW	foreign word	[gemeinschaft, hund, ich, jeux, habeas, Haemen…
13	IN	preposition or conjunction,	[astride, among, uppon, whether, out,

		subordinating	inside, …
14	JJ	adjective or numeral, ordinal	[third, ill-mannered, pre-war, regrettable, oi…
15	JJR	adjective, comparative	[bleaker, braver, breezier, briefer, brighter,…
16	JJS	adjective, superlative	[calmest, cheapest, choicest, classiest, clean…
17	LS	list item marker	[A, A., B, B., C, C., D, E, F, First, G, H, I,…
18	MD	modal auxiliary	[can, cannot, could, couldn’t, dare, may, migh…
19	NN	noun, common, singular or mass	[common-carrier, cabbage, knuckle-duster, Casi…
20	NNP	noun, proper, singular	[Motown, Venneboerger, Czestochwa, Ranzer, Con…
21	NNPS	noun, proper, plural	[Americans, Americas, Amharas, Amityvilles, Am…
22	NNS	noun, common, plural	[undergraduates, scotches, bric-a-brac, produc…
23	PDT	pre-determiner	[all, both, half, many, quite, such, sure, this]
24	POS	genitive marker	[‘, ‘s]
25	PRP	pronoun, personal	[hers, herself, him, himself, hisself, it, its…
26	PRP$	pronoun, possessive	[her, his, mine, my, our, ours, their, thy, your]
27	RB	adverb	[occasionally, unabatingly, maddeningly, adven…
28	RBR	adverb, comparative	[further, gloomier, grander, graver, greater, …
29	RBS	adverb, superlative	[best, biggest, bluntest, earliest, farthest, …
30	RP	particle	[aboard, about, across, along, apart, around, …
31	SYM	symbol	[%, &, ‘, ”, ”.,),)., *, +, ,., <, =, >, @…
32	TO	“to” as preposition or infinitive marker	[to]
33	UH	interjection	[Goodbye, Goody, Gosh, Wow, Jeepers, Jee-sus, …
34	VB	verb, base form	[ask, assemble, assess, assign, assume, atone,…

35	VBD	verb, past tense	[dipped, pleaded, swiped, regummed, soaked, ti...
36	VBG	verb, present participle or gerund	[telegraphing, stirring, focusing, angering, j...
37	VBN	verb, past participle	[multihulled, dilapidated, aerosolized, chaire...
38	VBP	verb, present tense, not 3rd person singular	[predominate, wrap, resort, sue, twist, spill,...
39	VBZ	verb, present tense, 3rd person singular	[bases, reconstructs, marks, mixes, displeases...
40	WDT	WH-determiner	[that, what, whatever, which, whichever]
41	WP	WH-pronoun	[that, what, whatever, whatsoever, which, who,...
42	WP$	WH-pronoun, possessive	[whose]
43	WRB	Wh-adverb	[how, however, whence, whenever, where, whereby...
44	“	opening quotation mark	[`, “]